

IMEN266 Operations Research II

Concept & Proof Companion — Chapters 1–3 (Basic Probability)

[GAP-FILL version — fill the blanks yourself first]

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How to use this document

This companion *replaces your class notes*: every definition and theorem from the Ch.1–3 slides is written out here in full, so you can watch and think in class instead of transcribing. It does **not** replace *doing*: for each core proof, (1) work through the GAP-FILL version with pen and paper, (2) check against the FULL version, (3) answer the Checkpoints, and (4) if stuck — or done — use an AI study card. Exams are closed-book; these six proofs and their checkpoints are exactly the kind of reasoning the exams reward.

1 Concept digest

1.1 Probability basics

A **sample space** Ω collects all outcomes of an experiment; an **event** is a subset $A \subseteq \Omega$. A probability measure P satisfies the axioms (i) $P(A) \geq 0$, (ii) $P(\Omega) = 1$, (iii) countable additivity: for *disjoint* $A_1, A_2, \dots$, $P(\bigcup_i A_i) = \sum_i P(A_i)$. Useful consequences: $P(A^c) = 1 - P(A)$; if $A \subseteq B$ then $P(A) \leq P(B)$; and for any two events

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \quad (\text{Proof 1}).$$

The general **inclusion–exclusion** law for $E_1, \dots, E_n$ alternates sums over single events, pairs, triples, ... (statement on slide p.3/35).

1.2 Conditional probability and independence

For $P(B) > 0$, $P(A \mid B) = \frac{P(A \cap B)}{P(B)}$ — “probability of A inside the world where B happened.” Events A, B are **independent** if $P(A \cap B) = P(A)P(B)$ (equivalently $P(A \mid B) = P(A)$). For three or more events, *mutual* independence requires the product rule for *every* sub-collection, not just pairs.

1.3 Reliability warm-up

With independent components, component i working with probability p_i :

$$R_{\text{series}} = \prod_{i=1}^n p_i, \quad R_{\text{parallel}} = 1 - \prod_{i=1}^n (1 - p_i).$$

(verified in notebook: CP1 warm-up) Chaining hurts, redundancy helps; Ch.9 generalizes both via structure functions.

1.4 Law of total probability and Bayes' rule

Let $B_1, \dots, B_n$ **partition** Ω (mutually exclusive, exhaustive, $P(B_i) > 0$). Then for any event A :

$$(\mathbf{LTP}) \quad P(A) = \sum_{i=1}^n P(A | B_i) P(B_i), \quad (\mathbf{Bayes}) \quad P(B_j | A) = \frac{P(A | B_j)P(B_j)}{\sum_i P(A | B_i)P(B_i)}.$$

(verified in notebook: CP1 #1–#3) Read Bayes as: *posterior* $\propto$ *prior* $\times$ *likelihood*.

1.5 Random variables and distribution functions

A **random variable** X assigns a number to each outcome. Its **cdf** $F(x) = P(X \leq x)$ always satisfies: non-decreasing; $\lim_{x \rightarrow -\infty} F = 0$, $\lim_{x \rightarrow \infty} F = 1$; right-continuous. *Discrete* X : pmf $p(x) = P(X = x) \geq 0$, $\sum_x p(x) = 1$. *Continuous* X : pdf $f \geq 0$, $\int f = 1$, $P(a < X < b) = \int_a^b f = F(b) - F(a)$, and $P(X = x) = 0$ for every point x .

1.6 Distribution zoo (as used in CP1–CP2)

Name	pmf / pdf	mean	variance	story (class problem)
Bernoulli(p)	$p^x(1-p)^{1-x}$	p	$p(1-p)$	one part: defective?
Binomial(n, p)	$\binom{n}{x} p^x q^{n-x}$	np	npq	#defects in 50 (CP1 #5a)
Geometric(p)	$q^{x-1}p, x \geq 1$	$1/p$	q/p^2	trial of 1st defect (#5b)
Neg. Binomial(r, p)	$\binom{x-1}{r-1} p^r q^{x-r}$	r/p	rq/p^2	trial of 3rd defect (#5c)
Poisson(λ)	$e^{-\lambda} \lambda^x / x!$	λ	λ	cars per minute (#6)
Hypergeom. (N, K, n)	$\binom{K}{x} \binom{N-K}{n-x} / \binom{N}{n}$	nK/N	—	lathes w/o replacement (#4)
Uniform(a, b)	$1/(b-a)$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	—
Exponential(λ)	$\lambda e^{-\lambda x}$	$1/\lambda$	$1/\lambda^2$	cell lifetime (CP2 #2)
Erlang(k, λ)	$\frac{\lambda^k x^{k-1} e^{-\lambda x}}{(k-1)!}$	k/λ	k/λ^2	message delay (CP2 #3)
Normal(μ, σ^2)	$\frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/2\sigma^2}$	μ	σ^2	—

Memorize the *stories*, not just the formulas: Binomial counts successes in a fixed number of trials; Geometric/Neg.-Binomial wait for the 1st/ r -th success; Poisson counts events at a rate; Hypergeometric samples *without* replacement; Erlang is k exponential phases in series.

1.7 Expectation

For discrete X : $E[X] = \sum_x x p(x)$; continuous: $E[X] = \int x f(x) dx$. **LOTUS** (Proof 5): $E[g(X)] = \sum_x g(x)p(x)$ or $\int g(x)f(x) dx$ — no need to find the distribution of $g(X)$. **Linearity** (Proof 4): $E[aX + bY] = aE[X] + bE[Y]$ *always*, with no independence assumption. **Variance**: $\text{Var}(X) = E[(X - \mu)^2] = E[X^2] - (E[X])^2$ (Proof 5); $\text{Var}(aX + b) = a^2 \text{Var}(X)$. **Tail-sum formula** (Proof 3): for $X \geq 0$, $E[X] = \int_0^\infty P(X > x) dx$ (continuous) and $E[X] = \sum_{k \geq 1} P(X \geq k)$ (nonnegative integer valued).

2 Core proofs

Notation: $q = 1 - p$. Blanks below are the steps *you* must supply in the gap-fill version.

Proof 1 — Inclusion–exclusion for two events

Theorem. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

Proof. Write $A \cup B$ as a union of *disjoint* pieces:

$$A \cup B = A \cup \underline{\hspace{2cm}}, \quad B = (B \cap A) \cup \underline{\hspace{2cm}}.$$

By additivity applied to each disjoint union,

$$P(A \cup B) = P(A) + P(B \cap A^c), \quad P(B) = \underline{\hspace{2cm}} + P(B \cap A^c).$$

Subtracting the second equation from the first eliminates $P(B \cap A^c)$ and yields $P(A \cup B) = P(A) + P(B) - \underline{\hspace{2cm}}$. $\square$

General case (idea). For $E_1, \dots, E_n$, apply expectation to the indicator identity $1 - \mathbf{1}_{\cup E_i} = \prod_i (1 - \mathbf{1}_{E_i})$ and expand the product: each k -fold term contributes $(-1)^k P(E_{i_1} \cap \dots \cap E_{i_k})$, giving the alternating sum on slide p. 3/35.

Checkpoint — answer before moving on.

- Where exactly did the proof use *disjointness*? Point to the line.
- Why does $P(A) + P(B)$ *overcount*, and what does the subtracted term count?
- Sanity check the general formula for $n = 3$ on a Venn diagram.

AI study card — paste one of these into your AI assistant:

- Quiz me step by step on the proof that $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. Ask one question at a time; if I'm wrong, give a hint, not the answer.
- Give me a concrete example where someone wrongly uses $P(A \cup B) = P(A) + P(B)$. Then make me fix it.

Proof 2 — Law of total probability and Bayes' rule

Theorem. If $B_1, \dots, B_n$ partition Ω and $P(B_i) > 0$, then $P(A) = \sum_i P(A \mid B_i)P(B_i)$ and $P(B_j \mid A) = P(A \mid B_j)P(B_j) / \sum_i P(A \mid B_i)P(B_i)$.

Proof. Since the B_i are exhaustive, $A = \bigcup_i (A \cap B_i)$, and since the B_i are mutually exclusive these pieces are $\underline{\hspace{2cm}}$. Additivity gives

$$P(A) = \sum_{i=1}^n P(A \cap B_i) = \sum_{i=1}^n \underline{\hspace{2cm}},$$

where the second equality is the multiplication rule $P(A \cap B_i) = P(A \mid B_i)P(B_i)$. For Bayes, start from the definition and apply the multiplication rule to the numerator and LTP to the denominator:

$$P(B_j \mid A) = \frac{P(B_j \cap A)}{P(A)} = \frac{\underline{\hspace{2cm}}}{\sum_i P(A \mid B_i)P(B_i)}. \quad \square$$

Checkpoint — answer before moving on.

- In CP1 #2 (cancer test), name B_1, B_2 and A explicitly, and match each factor of Bayes' formula to a number (.95, .04, .004, .996).
- "Posterior odds = prior odds $\times$ likelihood ratio." Derive this one-line form from Bayes for two hypotheses.
- What breaks in LTP if the B_i overlap? Give a 3-event example.

AI study card — paste one of these into your AI assistant:

- Play a skeptical student. I will explain the proof of the law of total probability; interrupt me whenever a step is not justified.
- Construct a base-rate-neglect trap problem different from the cancer test, let me solve it, then grade my solution harshly.

Proof 3 — Tail-sum formulas for expectation

Theorem. (a) If $X \in \{0, 1, 2, \dots\}$, then $E[X] = \sum_{k=1}^{\infty} P(X \geq k)$. (b) If $X \geq 0$ is continuous, then $E[X] = \int_0^{\infty} P(X > x) dx$.

Proof of (a).

$$\sum_{k=1}^{\infty} P(X \geq k) = \sum_{k=1}^{\infty} \sum_{j=k}^{\infty} P(X = j) \stackrel{(*)}{=} \sum_{j=1}^{\infty} \sum_{k=1}^j P(X = j) = \sum_{j=1}^{\infty} \underline{\hspace{2cm}} = E[X],$$

where $(*)$ swaps the order of summation: the pair (k, j) appears exactly when $\underline{\hspace{2cm}}$, so each $P(X = j)$ is counted $\underline{\hspace{1cm}}$ times. (Nonnegative terms make the swap legitimate.)

Proof of (b). Write $X = \int_0^{\infty} \mathbf{1}\{X > x\} dx$ (the integrand is 1 exactly while $x < X$). Taking expectations and interchanging E and $\int$ (Fubini–Tonelli, nonnegative integrand):

$$E[X] = \int_0^{\infty} E[\mathbf{1}\{X > x\}] dx = \int_0^{\infty} \underline{\hspace{2cm}} dx. \quad \square$$

Corollary (used constantly). For $X \sim \text{Geometric}(p)$: $P(X \geq k) = q^{k-1}$, so

$$E[X] = \sum_{k \geq 1} q^{k-1} = \frac{1}{1-q} = \underline{\hspace{1cm}}.$$

(verified in notebook: CP2 bridge cell: both tail integrals matched $E[X]$ numerically)

Checkpoint — answer before moving on.

- Draw the (k, j) lattice and shade the region summed in $(*)$ both ways.
- Use (b) to get $E[X] = 1/\lambda$ for the exponential in one line.
- Why does the theorem require $X \geq 0$? What goes wrong for a r.v. that takes negative values?

AI study card — paste one of these into your AI assistant:

- I will prove $E[X] = \sum_{k \geq 1} P(X \geq k)$ by swapping a double sum. Check each index bound I state and refuse to accept hand-waving.
- Give me three quick exercises that are one-liners with the tail-sum formula but painful without it.

Proof 4 — Linearity of expectation (no independence needed!)

Theorem. For any two random variables on the same space and constants a, b : $E[aX + bY] = aE[X] + bE[Y]$.

Proof (discrete case). Let $p(x, y)$ be the joint pmf. By LOTUS applied to $g(x, y) = ax + by$:

$$E[aX + bY] = \sum_x \sum_y (ax + by) p(x, y) = a \sum_x \sum_y x p(x, y) + b \sum_x \sum_y y p(x, y).$$

In the first double sum, sum over y first: $\sum_y p(x, y) = \underline{\hspace{1cm}}$ (the marginal), giving $a \sum_x x p_X(x) = \underline{\hspace{1cm}}$; symmetrically the second term is $bE[Y]$. At no point did we factor $p(x, y)$ — *independence was never used*. $\square$

Checkpoint — answer before moving on.

- Where *would* independence be needed: $E[XY] = E[X]E[Y]$ or $\text{Var}(X + Y) = \text{Var } X + \text{Var } Y$? Both?
- Use linearity + indicators to get the Binomial mean np in two lines (write $X = \sum_{i=1}^n \mathbf{1}_i$).
- CP1 #4: the lathes are sampled *without* replacement (dependent draws!), yet $E[X] = 2 \cdot \frac{4}{7}$. Which theorem just paid off?

AI study card — paste one of these into your AI assistant:

- Ask me to compute the expected number of fixed points of a random permutation using linearity and indicators; coach me only with questions.
- Try to convince me (wrongly) that linearity of expectation requires independence; I have to locate the flaw in your argument.

Proof 5 — LOTUS and the variance identity

Theorem (LOTUS, discrete). $E[g(X)] = \sum_x g(x) p_X(x)$.

Proof. Let $Y = g(X)$ and group outcomes by their y -value:

$$E[Y] = \sum_y y P(Y = y) = \sum_y y \sum_{x: g(x)=y} \underline{\hspace{1cm}} = \sum_y \sum_{x: g(x)=y} \underline{\hspace{1cm}} p_X(x) = \sum_x g(x) p_X(x),$$

since every x belongs to exactly one group $\{x : g(x) = y\}$. $\square$

Theorem (variance identity). $\text{Var}(X) = E[X^2] - (E[X])^2$.

Proof. Expand with $\mu = E[X]$ and use linearity (Proof 4):

$$E[(X - \mu)^2] = E[X^2 - 2\mu X + \mu^2] = E[X^2] - 2\mu \underline{\hspace{1cm}} + \mu^2 = E[X^2] - \underline{\hspace{1cm}}. \quad \square$$

(verified in notebook: CP2 #5: $E[X] = 3$, $E[X^2] = 12$, so $\text{Var} = 3$)

Checkpoint — answer before moving on.

- In LOTUS, which side of the identity requires knowing the distribution of $g(X)$, and why is avoiding that a big deal in practice?
- Constants slide out squared: prove $\text{Var}(aX + b) = a^2 \text{Var}(X)$ in two lines.

- CP2 #5 computed $E[X^2]$ by LOTUS. What integral would you have needed *without* LOTUS?

AI study card — paste one of these into your AI assistant:

- Generate a pdf for which $E[X]$ exists but $E[X^2]$ does not; make me verify both claims by integration.
- I claim $\text{Var}(X) = E[X^2] - (E[X])^2$ can be negative for some X . Argue with me until one of us proves the other wrong.

Proof 6 — The memoryless property (exam classic)

Theorem. (a) If $X \sim \text{Geometric}(p)$, then for integers $m, n \geq 0$: $P(X > m + n \mid X > m) = P(X > n)$. (b) If $X \sim \text{Exponential}(\lambda)$, then for $s, t \geq 0$: $P(X > s + t \mid X > s) = P(X > t)$.

Proof of (a). First, the survival function: $X > k$ means the first k trials were all failures, so $P(X > k) = _$ (no binomial coefficient — there is only one way to fail k times in a row). Then

$$P(X > m + n \mid X > m) = \frac{P(X > m + n, X > m)}{P(X > m)} = \frac{P(X > m + n)}{P(X > m)} = \frac{q^{m+n}}{_} = q^n = P(X > n),$$

where the second equality holds because $\{X > m + n\} \subseteq _$. □

Proof of (b). With $P(X > t) = e^{-\lambda t}$:

$$P(X > s + t \mid X > s) = \frac{P(X > s + t)}{P(X > s)} = \frac{e^{-\lambda(s+t)}}{_} = _ = P(X > t). \quad \square$$

Remark (uniqueness, starred). These are the *only* memoryless distributions in their classes: memorylessness forces the survival function to satisfy $g(s + t) = g(s)g(t)$, whose only monotone solutions are $g(t) = e^{-\lambda t}$ (continuous) and $g(k) = q^k$ (integer).

Checkpoint — answer before moving on.

- Why does $P(X > k) = q^k$ carry no $\binom{k}{\cdot}$ factor, while $P(X = k)$ -type binomial counts do?
- “A used exponential component is as good as new.” Say precisely what is being conditioned on, and what is claimed.
- Show Erlang(2, λ) is *not* memoryless: compare $P(X > 2/\lambda \mid X > 1/\lambda)$ with $P(X > 1/\lambda)$ numerically.

AI study card — paste one of these into your AI assistant:

- Quiz me one step at a time: derive the memoryless property for the geometric and the exponential. Then make me prove that no other continuous distribution has it.
- Claim (falsely) that the Erlang distribution is memoryless because it is built from exponentials. I must refute you with a computation.

Where to next. Notebook bridge cell (CP2, last section) → these proofs → Checkpoint quiz at the start of the next lecture → HW 1 Part A (closed-book style, no AI on the first attempt).